FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2023

TEST DATE: 27-03-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. D

Sol. When the source approaches the observer,

$$f_1 = f\left(\frac{v}{v - v_s}\right) = f\left(1 - \frac{v_s}{v}\right)^{-1} \approx f\left(1 + \frac{v_s}{v}\right)$$

or $$\left(\frac{f_1 - f}{f}\right) \times 100 = \frac{v_s}{v} \times 100 = 10 \quad \text{...(i)}$$

In the second case, when the source recedes from the observer,

$$f_2 = f\left(\frac{v}{v + v_s}\right) = f\left(1 + \frac{v_s}{v}\right)^{-1} \approx f\left(1 - \frac{v_s}{v}\right)$$

$\therefore$ $$\left(\frac{f_2 - f}{f}\right) \times 100 = -\frac{v_s}{v} \times 100 = -10$$ [from equation (i)]

2. A

Sol. $$F = -\frac{dU}{dr} = -\frac{k}{r}, \frac{k}{r} = \frac{mv^2}{r} \quad \text{..... (i)}$$

$$E_n = \frac{1}{2}mv^2 + k \ln r \quad \text{..... (ii)}$$

$$mvr = \frac{nh}{2\pi} \quad \text{..... (iii)}$$

For More Material Join: @JEEAdvanced_2024

Solving these $E_n = \frac{k}{2}\left(1+\ln\left(\frac{n^2h^2}{4\pi^2 mk}\right)\right)$

required ratio = $\frac{E_2 - E_1}{E_4 - E_2} = 1.$

3. C

Sol. Weight of wax in air = x g-wt
Weight of metal in water = y g-wt
Weight of wax and metal both in water = z g-wt
Weight of wax alone in water = (z – y) g-wt

Specific gravity of wax = $\frac{\text{weight of wax in air}}{\text{loss of weight of wax in water}}$

$= \frac{\text{weight of wax in air}}{\text{weight of wax in air} - \text{weight of wax in water}}$

$= \frac{x}{x-(z-y)}.$

4. D

Sol. Gravitational field is zero at a distance $\left(\frac{2r}{3}\right)$ from 4m.

$$V = \left(-\frac{Gm}{r/3}\right) + \left(-\frac{G(4m)}{2\frac{r}{3}}\right) = -\frac{9Gm}{r}$$

5. B

Sol. Potential energy of mass m when it is midway between masses M_1 and M_2

$$U = -\frac{GM_1 m}{d/2} - \frac{GM_2 m}{d/2} = -\frac{2Gm}{d}(M_1 + M_2)$$

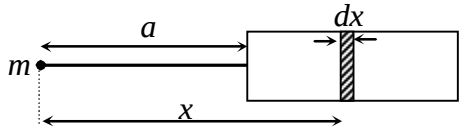
According to law of conservation of energy $\frac{1}{2}mv_e^2 = \frac{2Gm}{d}(M_1 + M_2)$

$\therefore$ v_e = escape velocity $= \sqrt{\frac{4G(M_1 + M_2)}{d}}$

6. B

Sol. $F = Gm\int_a^{a+L}\left(A + Bx^2\right)\frac{dx}{x^2}$

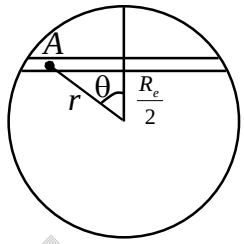
$$F = Gm\left[A\left[\frac{1}{a} - \frac{1}{a+L}\right] + BL\right]$$



For More Material Join: @JEEAdvanced_2024

7. D

Sol. Let mass of the earth is M_e and mass of object is m.



Force on object at $A = \left(\frac{GM_e m}{R^3} r\right)$ towards centre.

Force on object along the tunnel $= \left(\frac{GM_e m}{R^3} r\right) \frac{\sqrt{r^2 - \frac{R^2}{4}}}{r}$

Acceleration of object along the tunnel $= \left(\frac{GM_e}{R^3}\right) \sqrt{r^2 - \frac{R^2}{4}}$

8. A

Sol. Adiabatically

$P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$

$P_2 = P_1 \left(\frac{v_1}{v_2}\right)^{\gamma} = P_1 (\eta)^{\gamma}$ ...(1)

$\left[\text{as } \eta = \left(\frac{v_1}{v_2}\right)\right]$

Isothermally

$P_1 V_1 = P_2' V_2$

$P_2^{'} = P_1 \eta$ ...(2)

from (1) and (2)

$P_2 > P'_2$

9. B

Sol. In process $AB, \quad V \propto P$

$V = kP$

$PV = nRT$

$T = \frac{k}{nR} p^2$, parabolic

10. C

Sol. $\left(\frac{dm}{dt}\right) L = \frac{kA\theta}{x}$

$A\rho \frac{dx}{dt} L = \frac{kA[0-(-15)]}{x}$

$\int_{x_1}^{x_2} x\, dx = \int_0^t \frac{15kA}{A\rho L} dt$

$t = \frac{L}{30k}[x_2^2 - x_1^2]$

$\frac{t_2}{t_1} = \frac{[3.0^2 - 1.5^2]}{[1.5^2 - 0^2]} \Rightarrow t_2 = 20 \times 3 = 60 \text{ min}$

For More Material Join: @JEEAdvanced_2024

11. A

Sol. When switched is closed, inner plates of two capacitor get connected with each other. But outer plates are not connected. It means circuit is not completed. Therefore, no current will flow or no charge will appear on *B*.

12. A

Sol. It will only oscillate about the centre along the *z*-axis.

13. C

Sol. All the charge given to inner sphere will pass on to the outer one.
So, capacitor of that outer one = $4\pi\varepsilon_0 b$.

14. B

Sol. Resistance of cylinder $R = \int_0^l \frac{1}{\sigma}\frac{dx}{a} = \frac{2\sqrt{l}}{3a\sigma_0}$

$$I = \frac{E}{R}$$

Electric field $= \frac{J}{\sigma} = \frac{I}{a\sigma} = \frac{E\sqrt{x}}{Ra\sigma_0 l} = \frac{E(\sqrt{l})}{\frac{2\sqrt{l}}{3a\sigma_0}a\sigma_0 l} = \frac{3E}{2l}$

15. D

Sol. Magnetic flux linked with the loop is always zero.
So induced emf is also zero

16. A

Sol. Here required angles θ_1 and θ_2 are $(90-\phi_1)$ and $(90-\phi_2)$.

Hence $B = \frac{\mu_0 I}{4\pi d}\left[\sin(90-\phi_1)+\sin(90-\phi_2)\right] = \frac{\mu_0 I}{4\pi d}\left(\cos\phi_1+\cos\phi_2\right)$

17. A

Sol. By work energy theorem

$$mgz + BIlz - \int 2k(x+z)dz = 0$$

$2mgz = 2k\left[\int_0^z x\,dz + \int_0^z z\,dz\right]$ (where *x* is elongation in the equilibrium position)

$$2mg = mg + kz$$

$$z = \frac{mg}{k} = \frac{BIl}{k}$$

18. B

Sol. Conceptual

For More Material Join: @JEEAdvanced_2024

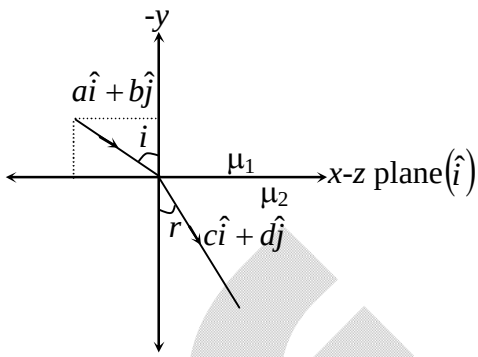
19. A

Sol. Applying Snell's law $\mu_1 \sin i = \mu_2 \sin r$

$$\left(\frac{3}{2}\right)\left(\frac{a}{\sqrt{a^2+b^2}}\right) = 2\left(\frac{c}{\sqrt{c^2+d^2}}\right)$$

Here $\sqrt{a^2+b^2} = \sqrt{c^2+d^2} = 1$

$\therefore \quad \frac{a}{c} = \frac{4}{3}$



20. A

Sol. We know that $N = N_0 (1/2)^{t/T}$ where N_0 = 16 g, t = 32 and T = 2

$N = 16\left(\frac{1}{2}\right)^{32/2} = 16\left(\frac{1}{2}\right)^{16}$ g, $N < 1$ mg

SECTION – B

21. 4

Sol. $T = 2\pi\sqrt{\frac{m}{K_{eq.}}}$

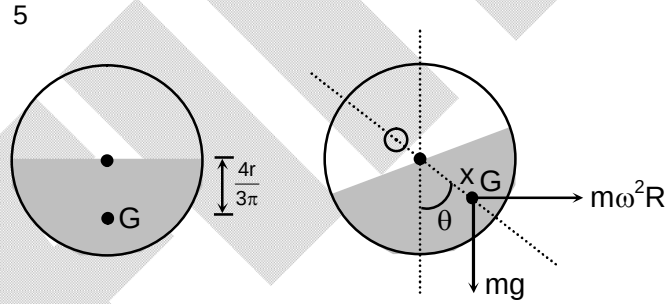
Where $K_{eq.} = \frac{1}{4\left[\frac{1}{K_1}+\frac{1}{K_2}+\frac{1}{K_3}+\frac{1}{K_4}\right]} = \frac{10}{16}$

$\therefore \quad T = 2\pi\sqrt{\frac{m}{K_{eq.}}} = 2\pi X$

$\therefore \quad X = \sqrt{\frac{10}{10/16}} = \sqrt{16} = 4.$

22. 5

Sol.



$I\alpha = -mgx\sin\theta$

$\Rightarrow \frac{1}{2}mr^2\frac{d^2\theta}{dt^2} = -mgx\sin\theta$

$\frac{d^2\theta}{dt^2} = -\frac{2g}{r^2}\left(\frac{4r}{3\pi}\right)\theta$

$\therefore \omega = \sqrt{\frac{8g}{3\pi r}} \Rightarrow f$

= 0.46 Hz

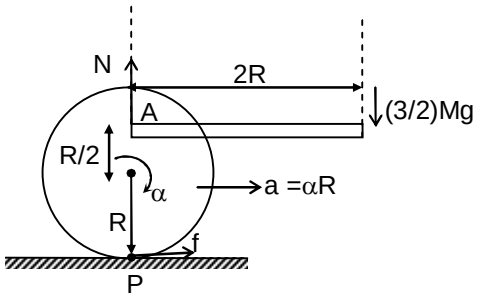
For More Material Join: @JEEAdvanced_2024

23. 3

Sol. Word-done = Area of P-V graph

24. 8

Sol.



$$\tau_{\text{about P}} = \frac{2R3Mg}{2} = \left(\frac{MR^2}{2} + MR^2\right)\alpha$$

$$3Mg = \frac{3MR}{2}\alpha$$

$$\frac{2g}{R} = \alpha$$

$$f = Ma = M\alpha R = 2Mg$$

$$N = Mg + \frac{3}{2}Mg = \frac{5Mg}{2}$$

$$\mu\frac{5Mg}{2} = 2Mg$$

$$\mu = \frac{4}{5} = 0.8$$

25. 35

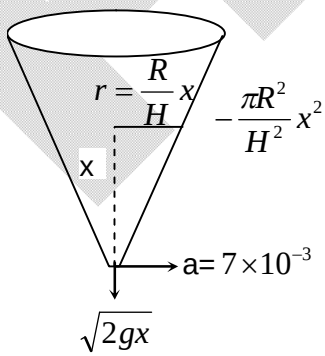
Sol. $T = 2t_1 + 2t_2$

$$= 2\left[\sqrt{\frac{2h}{g\sin^2\theta_1}} + \sqrt{\frac{2h}{g\sin^2\theta_2}}\right]$$

$$= 2\sqrt{\frac{2h}{g}}\left[\frac{1}{\sin\theta_1} + \frac{1}{\sin\theta_2}\right] = 35 \text{ sec}$$

26. 1

Sol.



$$-\frac{\pi R^2}{H^2}x^2\frac{dx}{dt} = \sqrt{2gx}a$$

For More Material Join: @JEEAdvanced_2024

$$\int_{H}^{0} x^{\frac{3}{2}} dx = -\sqrt{2g}\,\frac{H^2}{R^2}\,\frac{a}{\pi}\int_{0}^{t} dt$$

$$\frac{2}{5}H^{\frac{5}{2}} = \sqrt{2g}\,\frac{H^2}{R^2}\,\frac{a}{\pi}t \Rightarrow \frac{10^3}{5} = t \Rightarrow t = 200\,\text{sec}$$

27. 2

Sol. $$1700\left\{\frac{340}{340+V_2} - \frac{340}{340+V_1}\right\} = 10$$

$$\Rightarrow \quad \frac{1700\times 340}{340}\left\{\left(1-\frac{V_2}{340}\right)-\left(1-\frac{V_1}{340}\right)\right\} = 10$$

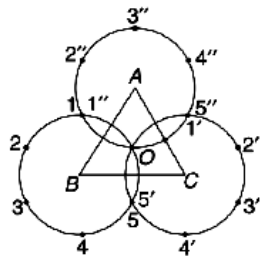
or $$\frac{1700(V_1 - V_2)}{340} = 10 \quad \text{or} \quad V_1 - V_2 = 2 \text{ m/s}$$

28. 12

Sol. Images formed due to multiple reflections in mirrors AB and BC will lie on a circle centred at B. Position of images are 1, 2, 3, 4, 5. Images due to reflection in BC and CA are 1′, 2′, 3′, 4′ and 5′. Images due to reflections in AB and AC are 1″, 2″, 3″ 4″ and 5″. 1 and 1″, 1′ and 5″ and 5′ coincide.

∴ Total number of images = 12.

Line joining the object and the image is perpendicular to the mirror, and object and the image are at equal distance from the mirror.



29. 2

Sol. Let h be the height to which the bullet rises then, $g^1 = g\left(1+\frac{h}{R}\right)^{-2}$

$$\Rightarrow \quad \frac{g}{4} = g\left(1+\frac{h}{R}\right)^{-2}$$

$$\Rightarrow \quad h = R$$

We know that $v_e = \sqrt{\frac{2GM}{R}} = v\sqrt{N}$ (given) ...(i)

Loss of kinetic energy = Gain in gravitational potential energy

$$\therefore \quad \frac{1}{2}mv^2 = -\frac{GMm}{2R} - \left(-\frac{GMm}{R}\right)$$

$$\therefore \quad v = \sqrt{\frac{GM}{R}}$$

Comparing (i) & (ii) N = 2

For More Material Join: @JEEAdvanced_2024

30. 4

Sol. Given, $V = kT^{2/3}$

$\therefore \quad dV = k \times \frac{2}{3} T^{-1/3} dT = \frac{2}{3} kT^{-1/3} dT$

Work done $dW = PdV$

$= \frac{RT}{V} dV$

$= \frac{RT}{kT^{2/3}} \times \frac{2}{3} kT^{-1/3} dT = \frac{2}{3} R(dT)$

Total work done $W = \frac{2}{3} R \int_{T_1}^{T_2} dT$

$= \frac{2}{3} R[T_2 - T_1]$

$= 2/3 \times 2 \times 30 = 40$ cal.

For More Material Join: @JEEAdvanced_2024

Chemistry

PART – B

SECTION – A

31. D

Sol. $C_6H_5NO_2 \xrightarrow{Fe/HCl} C_6H_5NH_2 \xrightarrow[+HCl,0^\circ C]{NaNO_2} C_6H_5\overset{\oplus}{N_2}\overset{\ominus}{Cl} \xrightarrow[\Delta]{H_2O/H^+} C_6H_5OH(\text{Phenol})$

32. B

Sol. $La^{+3} \rightarrow \text{Diamagnetic}[Xe]4f^o5d^o6s^o$

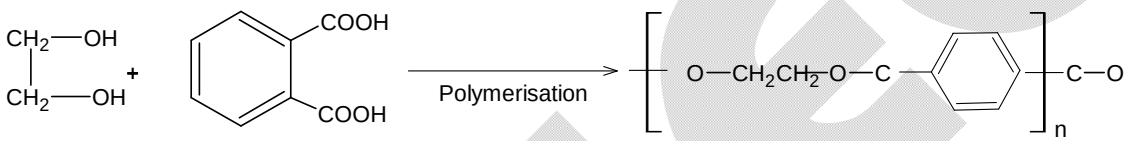
$Lu^{+3} \rightarrow \text{Diamagnetic}[Xe]4f^{14}5d^o6s^0$

$Gd^{+3} \rightarrow \text{Paramagnetic}[Xe]4f^7 5d^o 6s^o$

$Ce^{+4} \rightarrow$ Diamagnetic [Xe]$4f^o5d^o6s^o$

33. C

Sol.



34. C

Sol. $CO(g) + 2\,H_2(g) \xrightarrow[300^\circ C]{Cu/ZnO-Cr_2O_3} CH_3OH(g)$

35. B

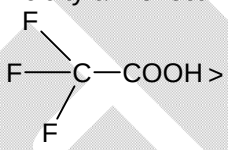
Sol. Bactericidal antibiotics:- antibiotics that kills the bacteria
Eg: Penicillin, Aminoglycosides, Ofloxacin
Bacteriostatic antibiotics: inhibitory effect on microbes
Eg: Erythromycin, Tetracycline, chloramphenicol

36. D

Sol. photochemical smog contains PAN, Ozone, Nitric oxide.

37. D

Sol. Acidity α–I effect



$> N \equiv C - CH_2COOH > HCOOH > CH_3 - COOH$

38. C

Sol. $Ni(OH)_2(aq) \rightleftharpoons \underset{S}{Ni^{+2}} + \underset{0.10M}{2^-OH}$

$NaOH \longrightarrow Na^+ + {}^-OH$

$0.10M \qquad\qquad 0.10M$

Ionic product = $2 \times 10^{-15} = S \times (0.10)^2$

$$S = \frac{2\times10^{-15}}{10^{-2}} = 2\times10^{-13}M$$

For More Material Join: @JEEAdvanced_2024

39. C

Sol. Emulsion:– Mixture 2 or more liquids, that show Brownian movement, show tyndall effect can be broken into constituents by centrifuging

40. D

Sol. BCC lattice, $Cs^+ \rightarrow$ body centre, $Br^- \rightarrow \frac{1}{8}\times 8 = 1$ corners

Unit cell length, a = 436.6 pm

Density of CsBr $= \frac{Z\times M}{a^3\times N_A}$

$$= \frac{1\times(133+80)}{\left(436.6\times10^{-10}\right)^3\times 6.022\times10^{23}}$$

$$= \frac{426}{85.15\times10^{-14}\times6.022\times10^{23}}$$

41. C

Sol. Thermite process: Reduction of metal oxides or non-metal oxides by Al and high temperature (heating).

$$M_xO_y + \frac{2}{3}yAl \longrightarrow x\,M + \frac{y}{3}Al_2O_3$$

42. B

Sol. rate = $k\,[A]^x[B]^y$

(I) $1.6\times10^{-3} = k[0.2]^x[0.2]^y$

(II) $1.6\times10^{-3} = k[0.2]^x[0.3]^y$

$$\left(\frac{0.2}{0.3}\right)^y = 1 \qquad \therefore\ y = 0$$

(III) $1.6\times10^{-3} = k[0.2]^x[0.2]^y$

(IV) $3.2\times10^{-3} = k[0.4]^x[0.2]^y$

$$\frac{1}{2} = \left(\frac{0.2}{0.4}\right)^x = \left(\frac{1}{2}\right)^x \quad x = 1$$

$\therefore$ r = k[A]

43. D

Sol. (I) $Fe^{+3} + 3e^- \xrightarrow[n_1=3]{} Fe,\ E_1^0 = 0.036\,V$

$$\Delta G_1^0, = -nFE_1^0 = +3\times F\times0.036 = 0.108F$$

(II) $Fe^{+2} + 2e^- \xrightarrow[n_2=2]{} Fe,\ E_2^0 = -0.44V$

$$\Delta G_2^0 = -n_2FE_2^\circ = +2\times F\times0.440$$

$$= +0.880F$$

$Fe^{+3} + e^- \longrightarrow Fe^{+2} \Rightarrow \Delta G^o = \Delta G_1^\circ - \Delta G_2^\circ$

$= -0.772\,F = -\,nFE_{cell}^\circ$

$\therefore\ E_{cell}^\circ = +0.772\,V$

For More Material Join: @JEEAdvanced_2024

44. B

Sol. $W_{rev.iso} > W_{rev.adia.} > W_{irr.adia}$

$\left[(T_f)_{gas}\right]_{rev.isothermal} > \left[(T_f)_{gas}\right]_{irr.adiabatic} > \left[(T_f)_{gas}\right]_{rev.adiabatic}$

45. B

Sol. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$K_p = P_{CO_2}$

$\Delta G^o = 130.2 \text{ kJ mol}^{-1} = -2.303RT \log K_p$

$$\log K_p = -\frac{130.2}{2.303 \times 0.0083 \times 700}$$

$= -9.73$

$K_p = \text{antilog } (-9.73) = \text{antilog } (0.29) \times 10^{-10}$

$P_{CO2} = K_p = 1.95 \times 10^{-10}$ atm

46. D

Sol. $\underset{\text{Sodium nitroprusside}}{Na_2\left[Fe(CN)_5NO\right]aq.} \xrightarrow{H_2S} \underset{\text{Violet/Purple colour solution}}{Na_4\left[Fe(CN)_5(NOS)\right]}$

47. A

Sol. (i) Basic strength of metallic oxide decreases across a period.

(ii) Alkaline and alkaline earth metal oxides are more basic than d-block oxides having incomplete octet.

48. B

Sol. SiO_2 has sp hybridisation of central atom and has linear structure.

49. C

Sol. $\left[Cr(OH)_4\right]^{\ominus} + H_2O_2 \xrightarrow{^-OH} CrO_4^{-2} + H_2O$

Oxidation Half

$\left[Cr(OH)_4\right]^{\ominus} \longrightarrow CrO_4^{-2} + 4H_2O$

$+4\,^-OH \quad +3e^-$

$\downarrow \times 2$

$6e^-$

Reduction Half

$H_2O_2 \longrightarrow 2H_2O + 2\,^-OH$

$+ 2H_2O$

$+2e^-$

$\downarrow \times 3$

$6e^-$

50. C

Sol. Order of reducing power

$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$

$\Delta EN \downarrow$ ses, Ionic character $\uparrow$ ses

SECTION – B

51. 3

Sol. According to spectrochemical series

Strong field ligands, pairs up d-electrons and can form low spin complexes.

Weak field ligands, do not pair electrons and will form high spin complexes.

$[FeF_6]^{-3}$ is outer orbital complex

For More Material Join: @JEEAdvanced_2024

52. 1
Sol. No. of radial nodes = n – *l* –1
For 4d orbital $\Rightarrow$ 4 –2–1 = 1

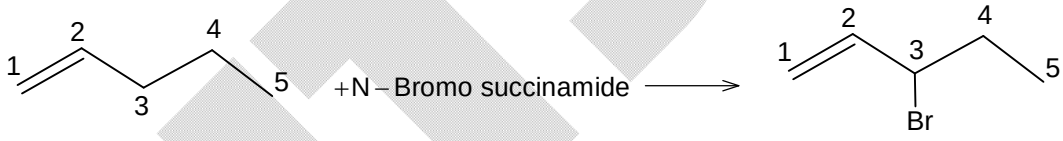
53. 3
Sol. For alkali metals, down the group (increasing atomic number)
Following properties increases:-
Metallic character, Ionic radius and density

54. 4
Sol. Ammonical $AgNO_3 \rightarrow$ Aldehyde test (silver mirror test)
$HIO_4 \rightarrow$ Vieinal – OH group
Alkaline $KMnO_4 \rightarrow$ Double bond
Chloroform + NaOH $\rightarrow$ Primary amine (Carbylamine reaction)

55. 3
Sol. $CH_3 - \underset{\underset{Br}{|}}{CH} - CH_2 - CH_3 \xrightarrow{\text{alc.KOH}} CH_2 = CH - CH_2CH_3 + CH_3 - CH = CH - CH_3$
cis & trans

56. 2
Sol. $Hg_2Cl_2 \longrightarrow Hg_2^{+2} + 2Cl^-$
$1-\alpha \qquad \alpha \qquad 2\alpha$
$\alpha = 50\%$
$i = 1 + 2\alpha = 1 + \left(2 \times \frac{1}{2}\right) = 2$

57. 3
Sol.



58. 6
Sol. 147% oleum = 100 gm + 47 gm
H_2SO_4 SO_3
Oleum + $H_2O \longrightarrow H_2SO_4 \xrightarrow{KOH}$ neutralisation
100 gm 47gm 147 gm
No. of equivalents of H_2SO_4 = No. of equivalent of KOH
$\frac{147}{98} \times 2 = x \times 500 \times 10^{-3}$
$x = \frac{147 \times 2 \times 10^3}{98 \times 500} = 6M$

59. 1
Sol. 1 mole of CO_2 at T = 273 K
Volume, V = 22.42 L
Van der Waal constant, a = 3.592 L^3 atm mol^{-2}
At 0ºC, 'b' be ignored as volume of molecules is negligible
$\therefore \quad \left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

For More Material Join: @JEEAdvanced_2024

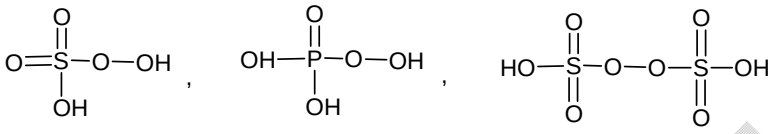
$$\Rightarrow \quad \left(P+\frac{a}{V^2}\right)(V)=RT$$

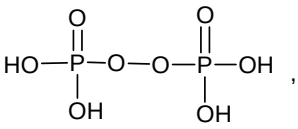
$$P=\frac{RT}{V}-\frac{a}{V^2}=\frac{RTV-a}{V^2}$$

$$P=\frac{(0.082\times 273\times 22.42)-3.592}{22.42\times 22.42}=0.99\,atm$$

$$P\sim 1\,atm$$

60. 4

Sol. 



4 structures having peroxy linkages

For More Material Join: @JEEAdvanced_2024

Mathematics

PART – C

SECTION – A

61. A

Sol. $\lim\limits_{x\to 0}\dfrac{\int_0^x \dfrac{t^2dt}{\sqrt{a+t}}}{(x-\sin x)} = \lim\limits_{x\to 0}\dfrac{x^2}{\sqrt{a+x}\,(1-\cos x)}$

$= \lim\limits_{x\to 0}\dfrac{x^2}{2\sin^2\dfrac{x}{2}.\sqrt{a+x}} = \dfrac{2}{\sqrt{a}} = 1$ (given)

$\Rightarrow$ a = 4.

62. D

Sol. Since they don't intersect each other

63. A

Sol. $2^x = \dfrac{a-3\pm\sqrt{(a-3)^2-4(a-4)}}{2} = a-4,\ 1$

It is given that roots are non-positive

i.e $x \le 0 \Rightarrow 2^x \le 1$

$\Rightarrow 0 < 2^x \le 1 \Rightarrow 4 < a \le 5.$

64. B

Sol. We have asinx +2bcosCcosx = α

The maximum value of asinx + 2bcosCcosx is $\sqrt{a^2+4b^2\cos^2 C}$ which increases as |cosC| increases. So the minimum value of |cosC| will correspond to the situation when $\sqrt{a^2+4b^2\cos^2 C}$ is just equal to α, so as to give a real solution of the equation.

$\therefore\ \sqrt{a^2+4b^2\cos^2 C} = \alpha$

$\cos^2 C = \dfrac{\alpha^2-a^2}{4b^2}$; $|\cos C| = \sqrt{\dfrac{\alpha^2-a^2}{4b^2}}$.

65. A

Sol. We have $(x_1+x_2+x_3+\ldots.+x_{50})\left(\dfrac{1}{x_1}+\dfrac{1}{x_2}+\ldots+\dfrac{1}{x_{50}}\right) \ge (50)^2$

[since A.M. $\ge$ H.M.]

$\Rightarrow \left(\dfrac{1}{x_1}+\dfrac{1}{x_2}+\ldots+\dfrac{1}{x_{50}}\right) \ge 50.$

66. B

Sol. $\left(1-x+\dfrac{x^2}{2!}-\dfrac{x^3}{3!}+\ldots.+\dfrac{(-1)^n x^n}{n!}\right)^2 = \left(e^{-x}-(-1)^{n+1}\dfrac{x^{n+1}}{(n+1)!}-(-1)^{n+2}\dfrac{x^{n+2}}{(n+2)!}+\ldots..\right)^2$

The required coefficient = coefficient of x^n in $e^{-2x} = \dfrac{(-2)^n}{n!}$.

For More Material Join: @JEEAdvanced_2024

67. D

Sol. $|iz + (3 - 4i)| \le |iz| + |3 - 4i|$

$= |z| + 5 < 4 + 5 = 9.$

68. B

Sol. Total number of permutations = $\frac{9!}{2!}$

Number of those containing 'HIN' = 7!

Number of those containing 'DUS' = $\frac{7!}{2!}$

Number of those containing 'TAN' = 7!

Number of those containing 'HIN' and 'DUS' = 5!

Number of those containing 'HIN' and 'TAN' = 5!

Number of those containing 'TAN' and 'DUS' = 5!

Number of those containing 'HIN', 'DUS' and 'TAN' = 3!

Required number = $\frac{9!}{2!} - \left(7! + 7! + \frac{7!}{2}\right) + 3 \times 5! - 3! = 169194.$

69. C

Sol. $0.7 = P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A).\ P(B)$

$= 0.2 + P(B) - 0.2P(B)$

$\Rightarrow \quad P(B) = \frac{0.5}{0.8} = \frac{5}{8}$

$\Rightarrow \quad P(B') = 3/8.$

70. A

Sol. $f(\theta) = \frac{(2\cos\theta - 1)(2\cos 2\theta - 1)(2\cos 4\theta - 1)\cdots(2\cos 2^{n-1}\theta - 1)}{2\cos 2^n\theta + 1}$ for $n \in N$

$= \frac{(2\cos\theta + 1)(2\cos\theta - 1)(2\cos 2\theta - 1)(2\cos 4\theta - 1)\cdots(2\cos 2^{n-1}\theta - 1)}{(2\cos\theta + 1)(2\cos 2^n\theta + 1)}$

$\cos\theta \ne -\frac{1}{2}$ as $\theta \ne 2m\pi \pm \frac{2\pi}{3}$ for $m \in I$.

$= \frac{(4\cos^2\theta - 1)(2\cos 2\theta - 1)\cdots(2\cos 2^{n-1}\theta - 1)}{(2\cos\theta + 1)(2\cos 2^n\theta + 1)}$

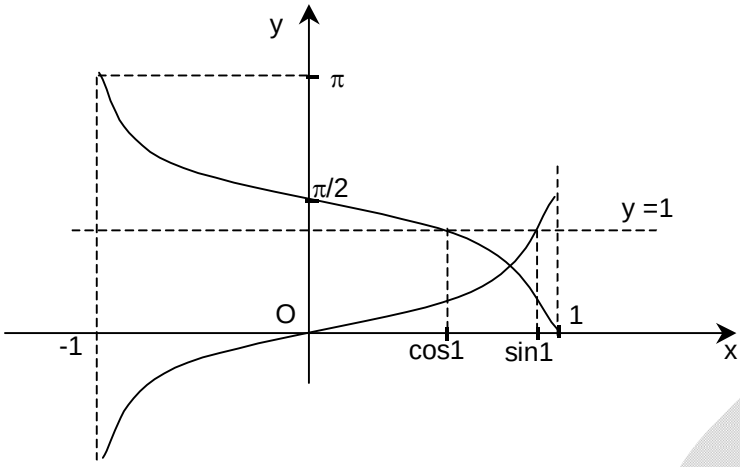
$= \frac{(2\cos 2\theta + 1)(2\cos 2\theta - 1)\cdots(2\cos 2^{n-1}\theta - 1)}{(2\cos\theta + 1)(2\cos 2^n\theta + 1)} = \frac{1}{2\cos\theta + 1}$

$\Rightarrow f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1.$`

For More Material Join: @JEEAdvanced_2024

71. A

Sol. $[\sin^{-1}x] > [\cos^{-1}x] \Rightarrow x > 0$



Clearly $[\cos^{-1}x] = \begin{cases} 1, & x \in [0, \cos 1] \\ 0, & x \in (\cos 1, 1] \end{cases}$

$[\sin^{-1}x] = \begin{cases} 0, & x \in [0, \sin 1) \\ 1, & x \in [\sin 1, 1] \end{cases}$

Hence $[\sin^{-1}x] > [\cos^{-1}x] \Rightarrow x \in [\sin 1, 1]$.

72. C

Sol. $\lim\limits_{x\to 0} \dfrac{2f(x) - 3f(2x) + f(4x)}{x^2}$ (Apply L'Hospital's Rule)

$\lim\limits_{x\to 0} \dfrac{2f'(x) - 6f'(2x) + 4f'(4x)}{2x} \Rightarrow \lim\limits_{x\to 0} \dfrac{2f''(x) - 12f''(2x) + 16f''(4x)}{2}$

$\dfrac{2.(4) - 12(4) + 16(4)}{2} = 12$

73. A

Sol. $f_p(\alpha) = \left(\cos\dfrac{\alpha}{p^2} + i\sin\dfrac{\alpha}{p^2}\right)\left(\cos\dfrac{2\alpha}{p^2} + i\sin\dfrac{2\alpha}{p^2}\right)\ldots.\left(\cos\dfrac{\alpha}{p} + i\sin\dfrac{\alpha}{p}\right).$

$= \left(\cos\dfrac{\alpha}{p^2} + i\sin\dfrac{\alpha}{p^2}\right)\left(\cos\dfrac{2\alpha}{p^2} + i\sin\dfrac{2\alpha}{p^2}\right)\ldots.\left(\cos\dfrac{p\alpha}{p^2} + i\sin\dfrac{p\alpha}{p^2}\right)$

$f_p(\alpha) = \cos\left(\dfrac{\alpha}{p^2} + \dfrac{2\alpha}{p^2} + \ldots. + \dfrac{p\alpha}{p^2}\right) + i\sin\left(\dfrac{\alpha}{p^2} + \dfrac{2\alpha}{p^2} + \ldots. + \dfrac{p\alpha}{p^2}\right)$

$= \cos\left\{\dfrac{\alpha}{p^2}\dfrac{(p)(p+1)}{2}\right\} + i\sin\left\{\dfrac{\alpha}{p^2}\dfrac{(p)(p+1)}{2}\right\}$

$f_n(2\pi) = \cos\dfrac{(n+1)\pi}{n} + i\sin\dfrac{(n+1)\pi}{n}$

$\lim\limits_{n\to\infty} f_n(2\pi) = \cos\left\{\lim\limits_{n\to\infty}\left(1+\dfrac{1}{n}\right)\pi\right\} + i\sin\left\{\lim\limits_{n\to\infty}\left(1+\dfrac{1}{n}\right)\pi\right\}$

$= \cos\pi + i\sin\pi = -1$

For More Material Join: @JEEAdvanced_2024

Alternate

$\lim_{n\to\infty}[\,|f_n(2\pi)|\,] = +1$ (not -1)

If $|\,|$ is not present

$\lim_{n\to\infty}[f_n(2\pi)] = [\lim_{n\to\infty} f_n(2\pi)]$ Assuming continuity

$[-1] = -1$.

74. C

Sol. Here $2k^2 + 4k - 16 = 2(k^2 + 2k - 8) > 0$

$\Rightarrow k \in (-\infty, -4) \cup (2, \infty)$.

75. A

Sol. $f(x) = \int_{5\pi/4}^{x} (3\sin t + 4\cos t)dt$, $x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$\Rightarrow f'(x) = 3\sin x + 4\cos x$

$f'(x) < 0$ as sinx, cosx are negative for $x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

$\Rightarrow f(x)|_{min} = f\left(\frac{4\pi}{3}\right) = \int_{5\pi/4}^{4\pi/3} (3\sin t + 4\cos t)dt = \frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$.

76. D

Sol. $f'(x) = (x^2-1)(2x) - (x-1) = (x-1)(2x^2 + 2x - 1)$, which is positive for $x > 1$. Hence f increases in [1,2]. Hence global maximum value of f is $f(2) = 4$

77. B

Sol. For $x \in (0, \infty)$, $[2e^{-x}] = 0$, except for $0 < x \le \ln 2$, in this case $[2e^{-x}] = 1$.

Thus $\int_0^{\infty} [2e^{-x}]dx = \int_0^{\ln 2} 1.dx = \ln 2$.

78. C

Sol. $I_1 = \int_0^{\pi/2} \frac{\cos^2 x}{1+\cos^2 x}dx$

$I_2 = \int_0^{\pi/2} \frac{\sin^2 x}{1+\sin^2 x}dx$

$I_3 = \int_0^{\pi/2} \frac{1+2\cos^2 x \sin^2 x}{4+2\cos^2 x \sin^2 x}dx$

Applying $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ in I_2

$\Rightarrow I_2 = \int_0^{\pi/2} \frac{\sin^2\left(\frac{\pi}{2}-x\right)}{1+\sin^2\left(\frac{\pi}{2}-x\right)}dx = \int_0^{\pi/2} \frac{\cos^2 x}{1+\cos^2 x}dx$

$\Rightarrow I_1 = I_2$

For More Material Join: @JEEAdvanced_2024

$$\therefore 2I_1 = \int_0^{\pi/2} \frac{\sin^2 x(1+\cos^2 x)+\cos^2 x(1+\sin^2 x)}{2+\sin^2 x\cos^2 x}dx$$

$$I_1 = \int_0^{\pi/2} \frac{1}{2}\left(\frac{1+2\sin^2 x\cos^2 x}{2+\sin^2 x\cos^2 x}\right)dx = I_3$$

$\Rightarrow I_1 = I_2 = I_3$.

79. A

Sol. Differential equation can be rewritten as

$$\left(\frac{xdy-ydx}{x^2}\right)\left(\frac{x^2}{y^2}\right) = (x\,dy + y\,dx)\sin xy$$

$$\text{or, } \left(d\left(\frac{y}{x}\right)\right)\left(\frac{x^2}{y^2}\right) = (d(xy))\sin xy$$

Integrating both sides, we get

$$-\frac{1}{y/x} = -\cos(xy) + c \Rightarrow \frac{x}{y} = \cos(xy) - c$$

Put x = 0, y =1, we get c = 1.

80. A

Sol. Let $\Delta(x) = A + Bx + Cx^2 + Dx^3 + \ldots$.

$$\Delta(0) = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 0 \Rightarrow A = 0$$

$$\Delta'(0) = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} = 1$$

$\Rightarrow \Delta(x) = x + Cx^2 + Dx^3 + \ldots\ldots \Rightarrow \Delta(x)$ is divisible by x

SECTION – B

81. 50

Sol. The common chord of the given circles is

$6x + 14y + l + m = 0$ (1)

Since $x^2 + y^2 + 4x + 22y + l = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - m = 0$, (1) passes through the centre of the second circle i.e. (1, -4).

Therefore, $6 - 56 + l + m = 0 \Rightarrow l + m = 50$.

82. 1

Sol. c =a/m

$\Rightarrow$ a = 2 (1/4) = 1/2

83. 12

Sol. A.M. $\geq$ G.M $\Rightarrow \frac{1}{2}(3\tan^2\theta + 12\cot^2\theta) \geq 6$

$\Rightarrow 3\tan^2\theta + 12\cot^2\theta$ has minimum value 12.

For More Material Join: @JEEAdvanced_2024

84. 0

Sol. We have $\sqrt{\frac{2-\sqrt{3}}{4}} = \sqrt{\frac{4-2\sqrt{3}}{8}} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

$\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} = \sin^{-1}\frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\pi}{12}$

$\cos^{-1}\frac{\sqrt{12}}{4} = \cos^{-1}\frac{\sqrt{3}}{2} = \pi/6$ and $\sec^{-1}\sqrt{2} = \pi/4$

We have $\sin^{-1}\left[\cot\left(\frac{\pi}{12}+\frac{\pi}{6}+\frac{\pi}{4}\right)\right] = \sin^{-1}\left(\cot\frac{\pi}{2}\right) = 0$.

85. 6

Sol. $S - S_n < \frac{1}{300}$

$\Rightarrow \frac{1}{1-\frac{1}{3}} - \frac{1-\frac{1}{3^n}}{1-\frac{1}{3}} < \frac{1}{300} \Rightarrow \frac{3}{2}\left[1-1+\frac{1}{3^n}\right] < \frac{1}{300}$

$\Rightarrow \frac{1}{3^n} < \frac{1}{300}\times\frac{2}{3} \Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$

$\therefore$ least value of n = 6

86. 1

Sol. Required area

$= \int_{\pi/4}^{3\pi/4} (\sin^2 x - \cos^2 x)\, dx$

$= \frac{1}{2}$

87. 25

Sol. $\vec{a}.\vec{b} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$

Thus we have to obtain the number of integral solution of this equation.

Coefficient of x° | $(x^{-3}+x^{-2}+x^{-1}+x^0+x+x^2)^3$

$= x^\circ \left| \left(\frac{1+x+x^2+x^3+x^4+x^5}{x^3}\right)^3 \right.$

$= x^9 |(1-x^6)^3 (1-x)^{-3} = {}^{11}C_9 - 3.{}^5C_3 = 25$.

88. 9

Sol.

$$S.D = \frac{\begin{vmatrix} 10 & 2 & 3 \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}}{\sqrt{8^2+8^2+4^2}} = \frac{108}{12} = 9$$

For More Material Join: @JEEAdvanced_2024

89. 7

Sol. Let L be the foot of the perpendicular drawn from the point P (1, 2, 3) to the given line the coordinate of a general point on

$$\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}=\lambda$$

$(x = 3\lambda + 6, y = 2\lambda + 7, z = -2\lambda + 7)$

Let these are the co–ordinate of the point L

so direction ratio's of P L is

$3\lambda + 6 - 1 , 2\lambda + 7 - 2 , -2\lambda + 7 - 3$

$\Rightarrow 3\lambda + 5 , 2\lambda + 5 , -2\lambda + 4$

and direction ratio of the given line is 3, 2, –2

Since PL is perpendicular to the given line, therefore

$3(3\lambda + 5) + 2(2\lambda + 5) + (-2) \times (-2\lambda + 4) = 0$

$\lambda = -1$

Putting value of λ in equation (1)

Hence L (3, 5, 9)

Hence PL = $\sqrt{(3-1)^2+(5-2)^2+(9-3)^2} = 7\text{unit}$

Hence required length of perpendicular is = 7 unit.

90. 0

Sol. $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$$\Delta = \begin{vmatrix} \sin(A+B)\sin(A-B) & \cot A - \cot B & 0 \\ \sin(B+C)\sin(B-C) & \cot B - \cot C & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \sin C \sin(A-B) & \dfrac{\sin(B-A)}{\sin A.\sin B} & 0 \\ \sin A \sin(B-C) & \dfrac{\sin(C-B)}{\sin B.\sin C} & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

Expanding along 3rd col. we have,

$$\Delta = \frac{\sin(A-B).\sin(C-B)}{\sin B} - \frac{\sin(B-C).\sin(B-A)}{\sin B}$$

$$\frac{\sin(A-B)}{\sin B}[-\sin(B-C)+\sin(B-C)] = 0$$

For More Material Join: @JEEAdvanced_2024